



PigeOff

Final Presentation

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Problem Description

- Monuments in Rome get defaced by pigeons
- Existing approaches:
 - Nets and spikes
 - Urban falconers
 - Optical gels
- Map pigeon activity through city

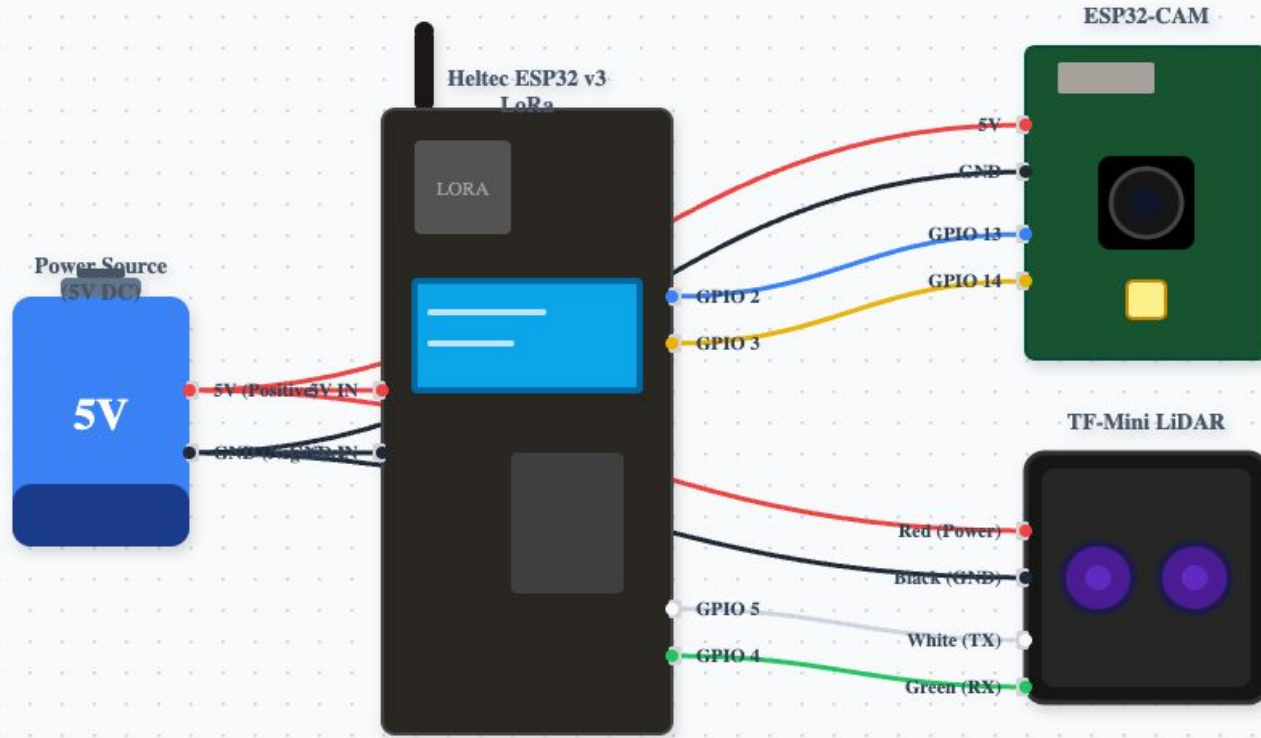


Overview

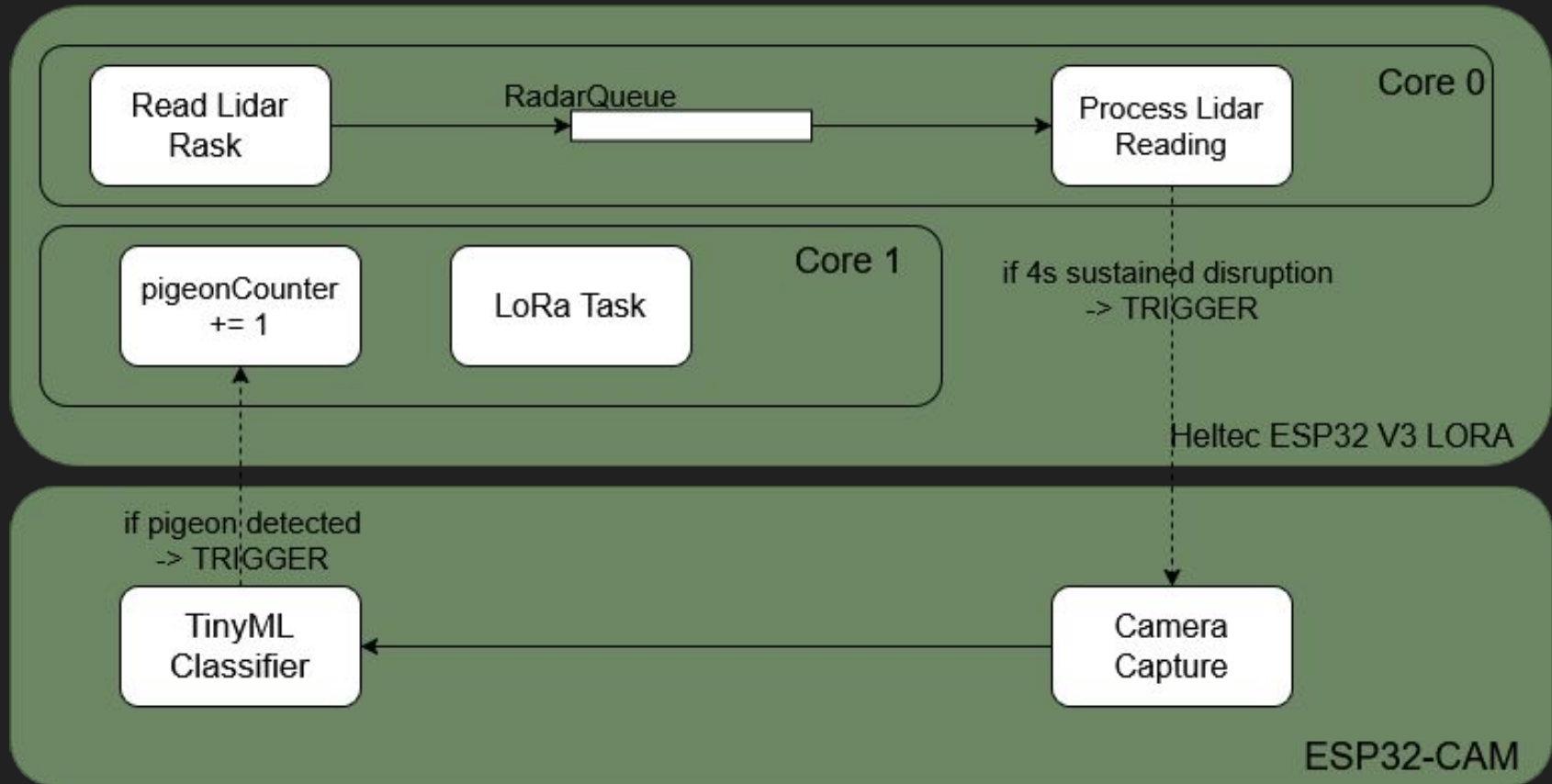
- Pigeon deterrent system
 - Radar/Lidar & Cam + LoRa
- Requirements
 - 7 day battery life
 - Feasible maintenance interval for municipality services
 - 10 second processing time
 - Pigeon average stay time 16.3 seconds
 - Defecating happens right before leaving



Architecture (System)



Architecture (FreeRTOS)



Architecture (FreeRTOS)

- Heltec enters light sleep between LiDAR samples (1s) unless the radio is actively transmitting
- ESP32-Cam in deep sleep unless LoRA board triggers wakeup

Classifier

1. Roboflow (Dataset) + Edge Impulse (MobileNetV1 96x96 0.25)
2. Handcrafted Dataset + Edge Impulse (Handcrafted model)



Classifier 1

Dataset size: 1740 – Train/test split: 80/20



INFERENCIN...
768 ms.



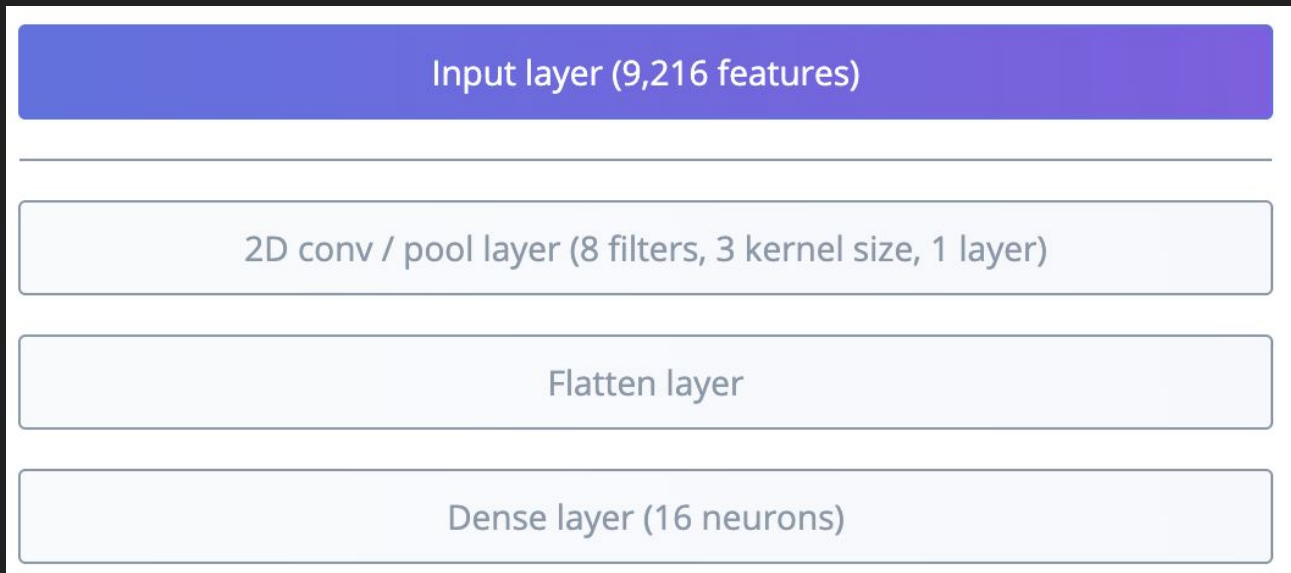
PEAK RAM U...
115.9K



FLASH USAGE
295.6K

Classifier 2

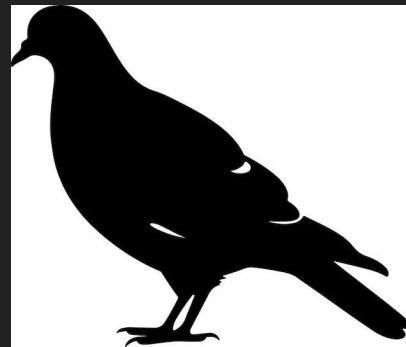
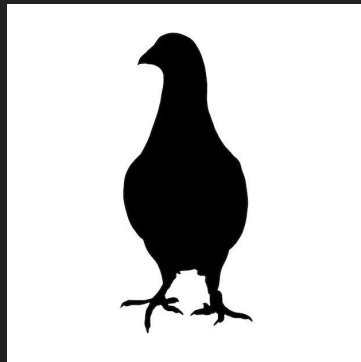
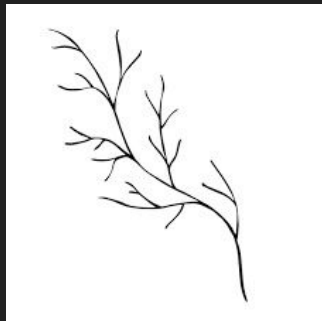
Dataset size: 60 – Train/test split: 80/20



Output layer (2 classes)

Classifier 2

Dataset size: 60 – Train/test split: 80/20



INFERENCE TIME
451 ms.



PEAK RAM USAGE
67.5K



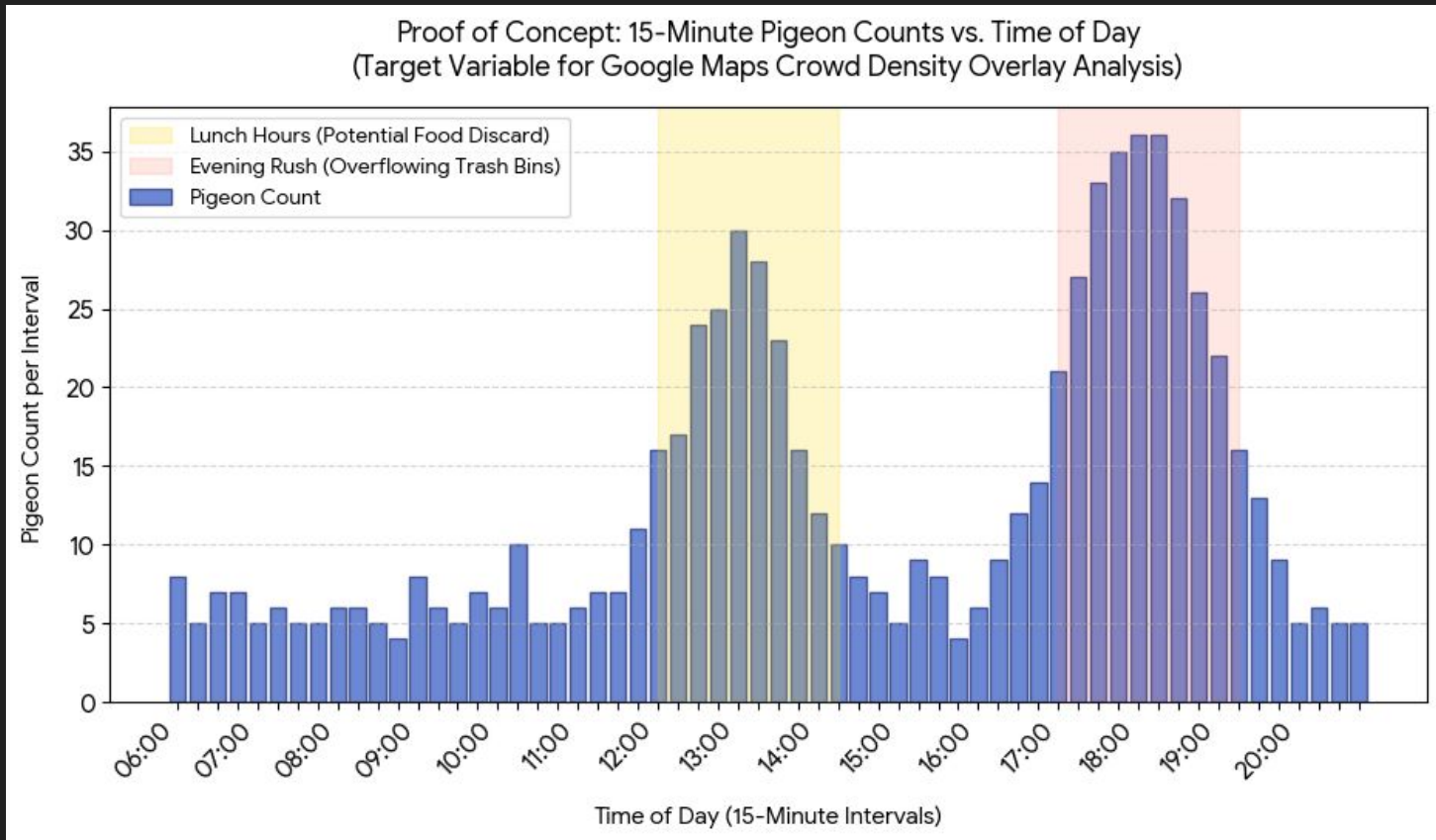
FLASH USAGE
329.8K



LoRa

- Unconfirmed uplinks to conserve energy
- In current location: SF7
- Pigeon count transmitted every 15 minutes

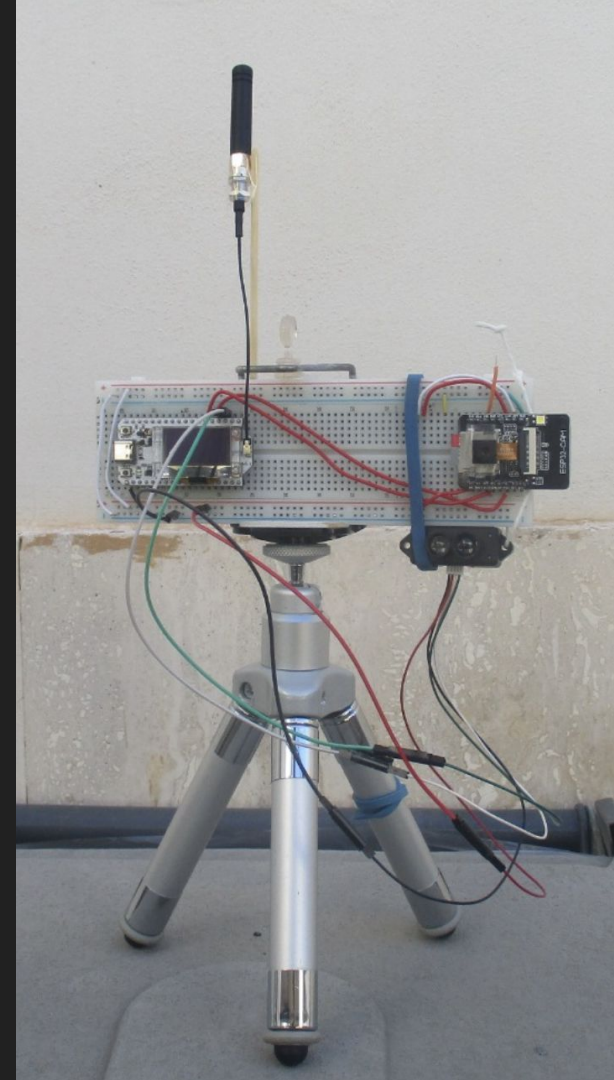
LoRa



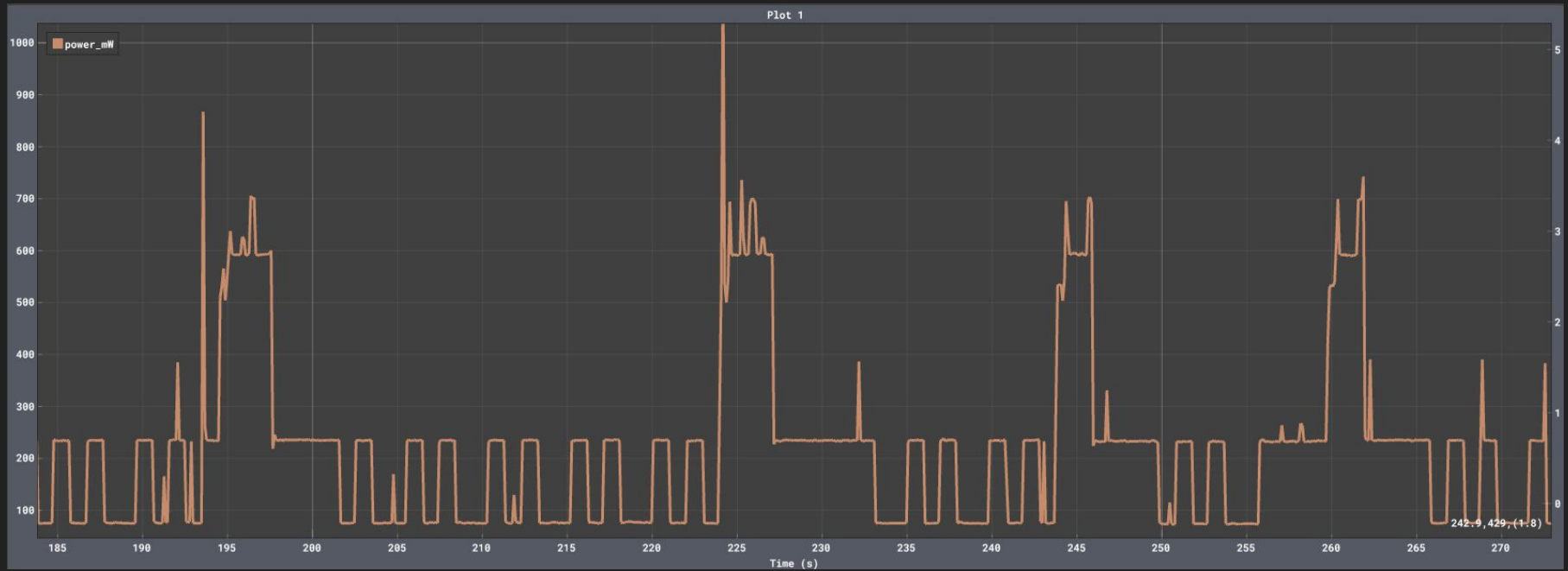
Evaluation

- Three distances: 20, 40, 80 cm
- Average inference time: 3000ms
- Optimal detection distance: 20cm

Distance	Lidar FN	Lidar FP	Camera FN	Camera FP	LiDAR F1	Camera F1
20	0%	20%	10%	0%	88.89%	94.74%
40	10%	0%	40%	0%	94.74%	75.00%
80	30%	0%	60%	0%	82.35%	57.14%



Energy Evaluation



Energy Evaluation

- We calculated an upper limit energy consumption
- Generalized as energy per hour
- Using averages from real power measurements
- Theoretical power consumption for the radar

Calculations for energy consumption

1 hour	3600	s
day	24	h
week	7	gg
freq birds/hor	26	
battery voltage	5	V

For the espcam

active delta t	3	s
energy active	0,34	mWh
power sleep	26,00	mW
<i>calculations over one hour of activityb</i>		
total energy active	8,84	mWh
total energy sleeping	25,44	mWh
TOTAL/hour	34,28	mWh

For the radar

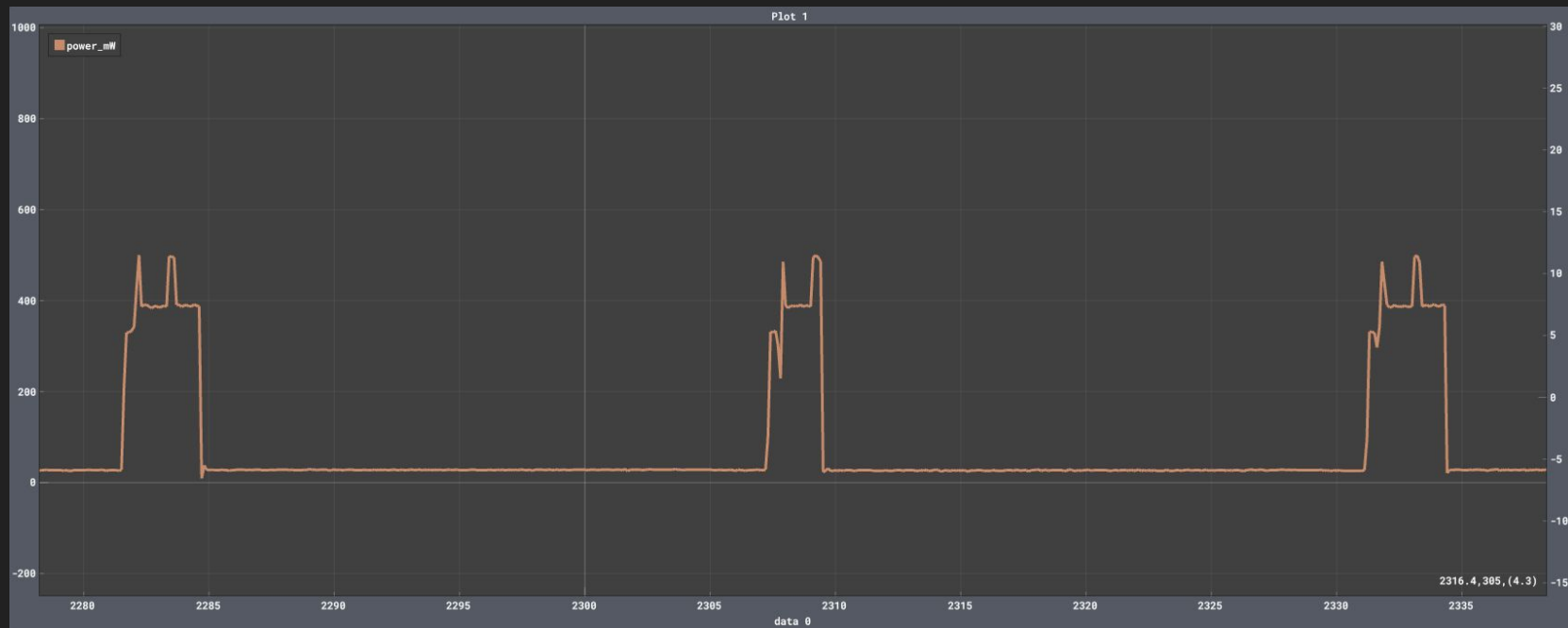
TOTAL/hour	0,396	mWh
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For the ESP32

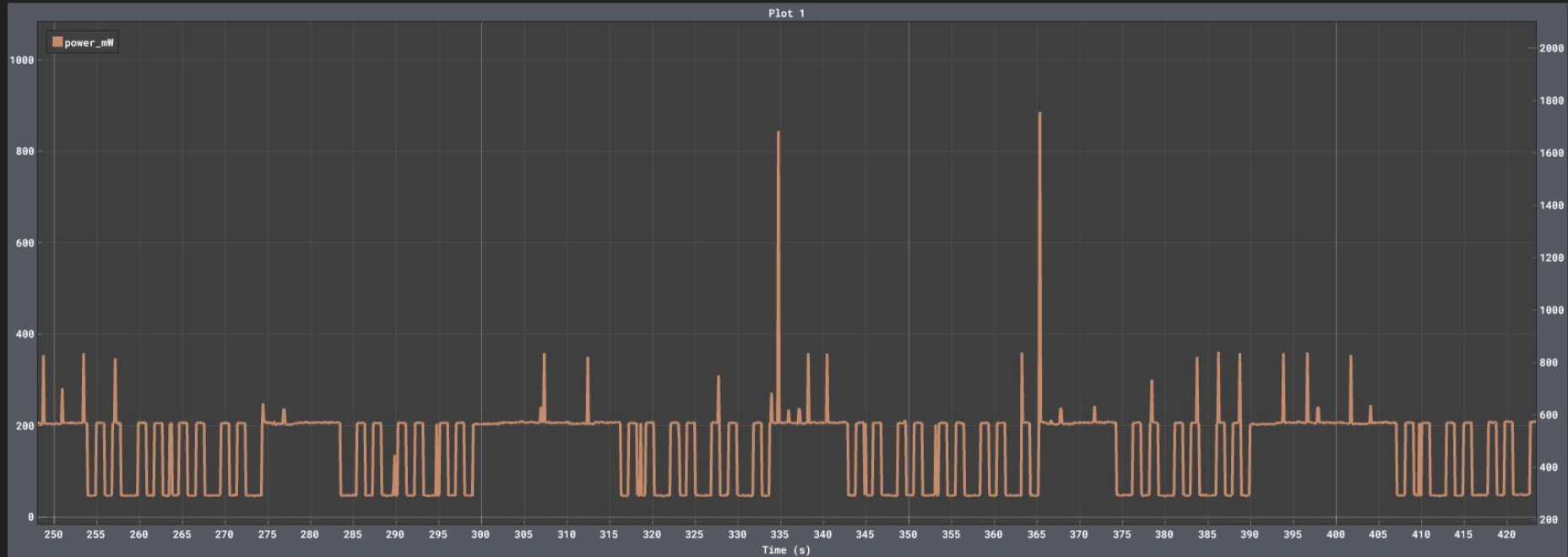
active delta t	9	s
energy active++	0,54	mWh
power sleep	46,00	mW
power active	204,00	mW
<i>calculations</i>		
total energy active++	14,04	mWh
total time not active ++	3366	s
ratio active remain time	0,53	
ratio sleeping remaining tim	0,47	
energy active	100,39	mWh
energy sleeping	20,37	mWh
TOTAL/hour	134,80	mWh
	169,48	mWh

Weekly energy	28.471,85	mWh
Battery capacity	5.694,37	mAh

Energy Evaluation (ESPCAM)



Energy Evaluation (ESP32)



Conclusions

- The average time for the inference on the ESPCAM is ~ 3s
- The average time for the ESP32 to operate is ~9s (<16s)
- Considering the worst case scenario of living 1 week always in peak activity time, the system can operate with a ~5700mAh battery

Further discussion

- Classifier too constrained?
- Better classification -> better features -> edge detection
- Sleeping cycle during inactivity periods (night)
- Implementation of device to scare the pigeons



DEMO

<https://youtu.be/5cjZkr73Wsk>



